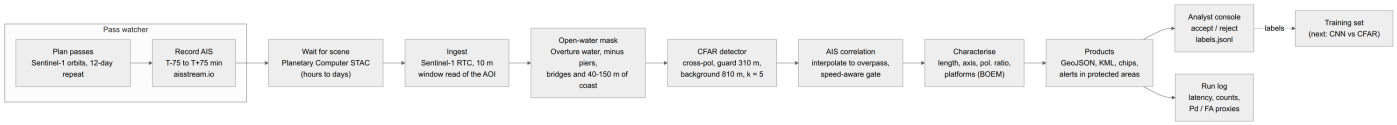
# Ships in Sentinel-1 radar, checked against AIS

## Operational-style ship detection and AIS correlation for maritime domain awareness

A scheduled service that detects ships in Sentinel-1 radar, decides which ones AIS accounts for, and flags the rest for an analyst. It is scored the way an operator would use it: probability of detection (Pd) at a fixed false-alarm rate. The validation uses **12** Sentinel-1 passes from 2024 over US waters, where complete AIS is public (NOAA MarineCadastre, CC0), plus a locked set of **36** passes in three areas never used for any choice. Production runs the classical **CFAR baseline**; a learned two-stage detector beats it at strict false-alarm budgets (Pd **0.32** vs **0.22** at 1 FA, averaged over training seeds) and ties at 5 FA and at production's false-alarm rate, small craft included, but failed the promotion gate. A second one (v6, a Sentinel-1 ResNet-50) passes the gate, but a pre-registered test in three new areas found it worse than CFAR at equal recall; the gate now checks that. Built by Shivaji Patil in October 2026, with an AI coding agent writing much of the code under my direction; every number is reproduced by scripts and checked by tests.

## WHAT RUNS

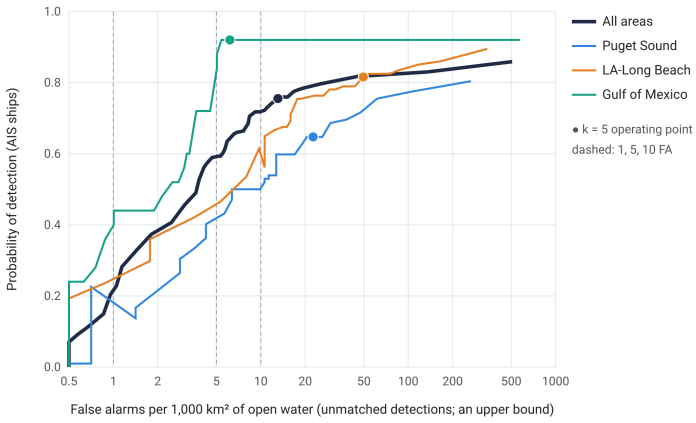



Timers record live AIS around each Sentinel-1 pass of Vancouver and St. John's. When the scene is published, it is masked to open water and searched with CFAR. Targets are correlated with AIS, characterised and turned into products and alerts, then reviewed in the analyst console.

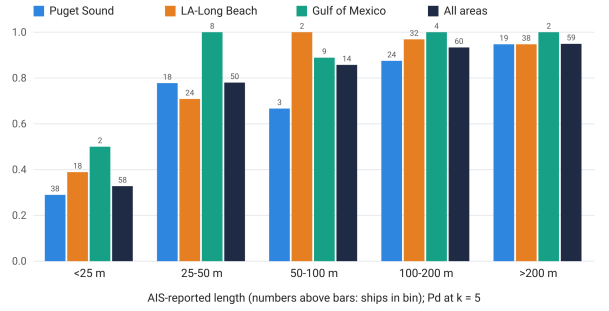
## VALIDATION: 12 PASSES IN 2024, NOAA AIS AS TRUTH

| Area | Passes | AIS ships | Water km$^2$ | Pd at 1 / 5 / 10 FA | Pd, k=5 | FA, k=5 | Unmatched | Platforms |
|---|---|---|---|---|---|---|---|---|
| **All areas** | 12 | 226 | 10,469 | **0.22** / **0.63** / **0.77** | **0.81** | **13.1** | 137 | 120 |
| Puget Sound | 4 | 92 | 1,412 | 0.25 / 0.45 / 0.55 | 0.72 | 22.7 | 32 | 0 |
| LA–Long Beach | 4 | 109 | 1,125 | 0.25 / 0.44 / 0.64 | 0.84 | 50.7 | 57 | 0 |
| Gulf of Mexico | 4 | 25 | 7,933 | 0.36 / 0.92 / 0.96 | 0.96 | 6.1 | 48 | 120 |

AIS ships on open water at least 150 m from the mapped coast. FA: detections that no AIS ship accounts for, per 1,000 km$^2$ of open water. This is an upper bound, because vessels without AIS count against it. Platforms: detections within 150 m of a standing BOEM platform, left out of the scores. A match must lie within 300 m of a region stretched along the satellite track by a moving ship's expected image shift; random detections would reach Pd **0.04**.



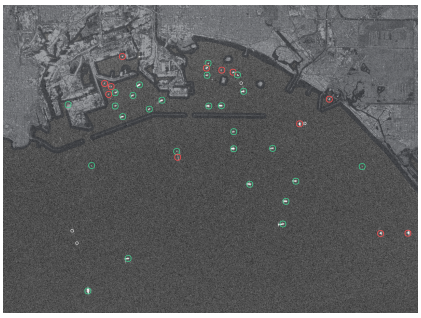

Operating curves. The detector is re-run at each threshold, so every point is a real fixed-threshold detector. Dots mark k = 5.



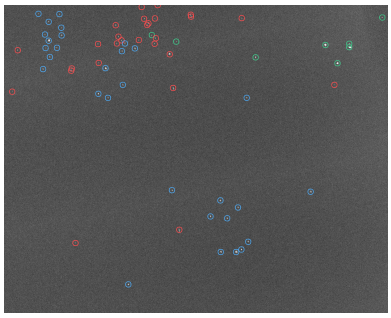

Pd at k = 5 by AIS length, all areas: **0.36** under 25 m, **0.83** at 25–50 m and **0.95**–**0.98** over 100 m. Small craft are often below what 10 m pixels resolve.

- **Position error:** stationary ships median **42 m**, p90 **139 m**. Moving ships median **152 m**, p90 634 m: a moving ship is imaged shifted along the track by about 115 s times its radial speed.
- **Locked test** (Massachusetts Bay, Chesapeake entrance, New York Lower Bay; 36 passes, 284 ships, list committed before any data was fetched): Pd **0.69** at k = 5, **0.94** for ships of 50 m and more, **13.2** FA. CEP90 for stationary ships is 63 m, and a 90% position ellipse on every detection holds **90%** of AIS positions. Its first scoring exposed a truth fault (AIS clipped to the box), since fixed for every set; the detector did not change.

## QUALITY CONTROL AND WHAT IT CHANGED



LA–Long Beach, 20 Apr 2024: **green** matched to AIS, **red** unmatched; white rings are AIS ships not detected.

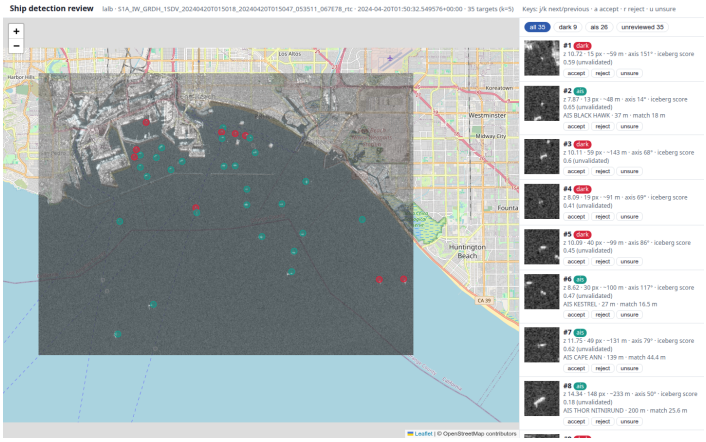


Gulf of Mexico platforms, 12 Oct 2024: **blue** within 150 m of a BOEM platform (30 in this pass), kept out of false alarms.

- **Coastal buffer.** With Harbour Twin's 40 m coastal buffer, the QC images showed Puget Sound's unmatched targets lined up along the shore. They were private docks and shore returns the map does not show: **350** unmatched at 40 m against **32** at 150 m. The cost is that ships within 150 m of the coast are not scored (272 AIS ships at 40 m, 226 at 150 m). Re-chosen on the 23 training passes alone, the rule picks 300 m: on the test passes Pd **0.66** at 5 FA (150 m: **0.63**), but only 201 of 226 ships scored. So 150 m is not flattered, and Pd alone cannot pick the buffer.

## CHARACTERISATION AND ANALYST REVIEW

- **Length and axis** come from the second moments of each target's pixels. Against AIS for **182** matched ships, radar length runs long, with a median bias of +9 m and a median error of 16 m (correlation **0.84**). The axis is within 20° of the AIS heading for **77%** of ships; the median error is 4°.
- **Platforms:** **120** Gulf detections fall at standing BOEM structures and would otherwise count as dark vessels.
- **Iceberg vs ship (St. John's):** a rule-based score from polarisation ratio, compactness and size. It is **not validated**: no official sightings with clear terms overlap these scenes.
- **Products per pass:** GeoJSON and KML, 640 m image chips, alerts for unmatched targets inside marine protected areas (CPCAD), and a run log with latency. Each pass is also written in the Harbour Twin viewer's format.




Analyst console on 127.0.0.1: map, target list with chips, accept or reject. Decisions are saved as labels. Base map © OpenStreetMap contributors.

## SECOND STUDY: DANISH WATERS AT SCALE

- **Physics-based matching:** on 2,585 fresh AIS matches, the along-track offset follows the Doppler azimuth shift (slopes **0.95** / **0.94**, theory 1.0); correcting it cuts the median offset from about 200 m to 60–65 m.
- **VH beats VV:** over 70 frames, the unmatched list halves (27.1 to 13.6 per 1,000 km$^2$) at similar recall.
- **Generalisation:** recall 81% in the straits, 92% in the North Sea, and 72% on 94 frames from a held-out month (May).
- **Foundation models:** on 2,739 AIS-labelled chips, a fine-tuned Sentinel-1 SoftCon backbone gives vessel-type macro-F1 **0.52** (ImageNet **0.48**) and length error 19–27 m for learned models, against 44 m for CFAR.

## OPERATIONS, LIMITS AND NEXT STEPS

- **Live:** first pass, Vancouver 22 Oct: 46 targets, 27 of 33 AIS ships found, 52 s to process, 12 h after sensing (Planetary Computer's publishing delay; near-real-time needs Copernicus NRT). St. John's passes follow on 26–28 Oct. Changes reach the service only through an MLflow registry and promotion gate, and each pass records the detector version.
- **Limits:** AIS is incomplete truth; boats under 25 m are often missed at 10 m; ships within 150 m of the coast are not scored; 48 passes, all US waters in 2024.

1. **Detector:** a false-alarm budget did not transfer (2.3x over); a structure map from each region's own history then cut dark-list false alarms 41% in five new areas (pre-registered); next, live use with analyst review of its spots.
2. **Discrimination:** azimuth ambiguities and sidelobes; icebergs with labelled sightings.
3. **Then:** AIS mismatch flags, cross-pass linking; near-real-time data, offline deployment, RCM.

Data: contains modified Copernicus Sentinel data (2024, 2026), Sentinel-1 RTC via Microsoft Planetary Computer. AIS: Martin, Brass, Dornback and Fontenault (2025), Nationwide AIS 2024, NOAA Office for Coastal Management, CC0. Live AIS: aisstream.io. Water and piers: Overture Maps Foundation (includes OpenStreetMap data, ODbL). Platforms: BOEM Platform Structures (U.S. Government). Protected areas: CPCAD (ECCC), Open Government Licence – Canada. Danish study: Danish Maritime Authority AIS; SSL4EO-S12 backbones. Related: Harbour Twin (GitHub).